

CS 3210 HW1

1.6

P1

Clock rate = 2.5 GHz

CPI(A,B,C,D) = (1,2,3,3)

P2

Clock rate = 3 GHz

CPI(A,B,C,D) = (2,2,2,2)

Total instructions = 1×10^6

Class	Division
A	$10\% = 10^6 / 10^2 = 10^4$ instructions
B	$20\% = 2 \times 10^4$ instructions
C	$50\% = 5 \times 10^4$ instructions
D	$20\% = 2 \times 10^4$ instructions

a) $CPI_{\text{global}} = \sum f_i * CPI_i$

For $P_1 = (0.01)(1) + (0.02)(2) + (0.5)(3) + (0.2)(3) = 0.01 + 0.04 + 1.5 + 0.6 = 2.6$

For $P_2 = (0.1)(2) + (0.2)(2) + (0.5)(2) + (0.2)(2) = 0.2 + 0.4 + 1 + 0.4 = 2$

b) Clock Cycles for $P_1 = 1 \times 10^6 \times 2.6 = 2.6 \times 10^6$

Clock Cycles for $P_2 = 1 \times 10^6 \times 2 = 2 \times 10^6$

Now, in order to determine which one is faster we need to calculate the ratio of the execution times of P_1 and P_2 .

Execution time of $P_1 = (\text{Clock cycles} / \text{clock rate}) = 2.6 \times 10^6 / 2.5 \times 10^9 = 1.04 \text{ ms}$

Execution time of $P_2 = 10^6 \times 2 / 3 \times 10^9 = 0.67 \text{ ms}$

Now, let us divide the execution time of P_1 by the execution time of P_2 .

$$= 1.04/0.67 = 1.56$$

Hence, we can say that P_2 is 1.56 times faster than P_1 .

1.8.1

Pentium 4 Prescott

$f = 3.6 \text{ GHz}$

$V = 1.25 \text{ V}$

$P_{\text{static}} = 10 \text{ W}$

$P_{\text{dynamic}} = 90 \text{ W}$

$$P_{\text{dynamic}} = \frac{1}{2} * C * V^2 * f$$

$$(2 * P_{\text{dynamic}}) / V^2 * f = C$$

$$(2 * 90) / (1.25)^2 * (3.6 * 10^9) = 3.2 \times 10^{-8} \text{ F}$$

Core I₅ Ivy Bridge

$f = 3.4 \text{ GHz}$

$V = 0.9 \text{ V}$

$P_{\text{static}} = 30 \text{ W}$

$P_{\text{dynamic}} = 40 \text{ W}$

$$P_{\text{dynamic}} = \frac{1}{2} * C * V^2 * f$$

$$(2 * P_{\text{dynamic}}) / V^2 * f = C$$

$$(2 * 40) / (0.9)^2 * (3.4 \times 10^9) = 2.9 \times 10^{-9} \text{ F}$$

1.8.2

Pentium 4 Prescott

$$P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}}$$

$$P_{\text{total}} = 10 + 90 = 100 \text{ W}$$

$$\text{Percentage of static power} = 10/100 * 100 = 10\%$$

$$\text{Ratio of static power to dynamic power} = 10/90 = 0.111$$

Core I₅ Ivy Bridge

$$P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}}$$

$$P_{\text{total}} = 30 + 40 = 70 \text{ W}$$

$$\text{Percentage of static power} = 30/70 * 100 = 42.9\%$$

$$\text{Ratio of static power to dynamic power} = 30/40 = 0.75$$

1.8.3

Let us assume that the old power dissipated was P_1 , and the new power dissipated is P_2 . Now, it is given that the new dissipated power was reduced by 10%, so we can say that $P_2 = (1 - 0.1) P_1 = 0.9 P_1$.

Hence, we know that the total power dissipated can be calculated as follows:

$$P_2 = \text{new static power}(I_2 V_2) + \text{new dynamic power}((\frac{1}{2}) * C_2 V_2^2 f_2) = 0.9 P_1$$

$$I_2 = 10/1.25 = 8 \text{ A}$$

Pentium Prescott Processor

$$8V_2 + (\frac{1}{2}) * 32 * 10^{-9} * V_2^2 * 3.6 * 10^9 = 0.9 * 100$$

$$8V_2 + 57.6 * V_2^2 = 90$$

$$57.6 * V_2^2 + 8V_2 - 90 = 0$$

After solving the quadratic equation, we get V_2 as 1.18V.

$$\text{Change in total dissipated power} = (1.18 - 1.25)/1.18 * 100 = -5.9 \%$$

Ivy Processor

$$P_2 = \text{new static power}(I_2 V_2) + \text{new dynamic power}((\frac{1}{2}) * C_2 V_2^2 f_2) = 0.9 P_1$$

$$I_2 = 30/0.9 = 33.333 \text{ A}$$

$$33.3 * V_2 + \frac{1}{2} * (2.9 * 10^{-9}) * V_2^2 * 3.4 * 10^4 = 0.9 * 70$$

$$33.33V_2 + 49.385V_2^2 = 63$$

$$33.33V_2 + 49.385V_2^2 - 63 = 0$$

After solving this quadratic equation, we get V_2 as 0.82V.

$$\text{Change in dissipated power} = (0.82-0.9)/0.82 * 100 = -9.756 \%$$

Hence, we can say that the percentage of change in dissipated powers is 5.9% and 9.8% for Pentium Prescott Processor and Ivy Processor.

1.10.1

$$\text{yield} = 1/(1+(\text{defects per unit area} * \text{dies per unit area}/2))^2$$

Wafer 1:

$$\text{Radius} = 15/2 = 7.5 \text{ cm}$$

$$\text{Total Area} = \pi * \text{radius} * \text{radius} = \pi * (7.5) * (7.5) = 176.71 \text{ cm}^2$$

$$\text{Area per die} = 176.71/84 = 2.1 \text{ cm}^2$$

$$y_1 (\text{or yield 1}) = 1/(1+(0.020*2.1)/2)^2 = 0.959$$

Wafer 2:

$$\text{Radius} = 20/2 = 10 \text{ cm}$$

$$\text{Total Area} = \pi * \text{radius} * \text{radius} = \pi * 10 * 10 = 314.159 \text{ cm}^2$$

$$\text{Area per die} = 314.159/100 = 3.14 \text{ cm}^2$$

$$y_2 (\text{or yield 2}) = 1/(1+(0.031*3.14)/2)^2 = 0.909$$

1.10.2

Formula for calculating the cost: cost per wafer/(Dies per wafer * yield)

$$c_1(\text{cost for wafer 1}) = 12/(84*0.959) = 0.148$$

$$c_2(\text{cost for wafer 2}) = 15/(100*0.909) = 0.165$$

1.10.3

Wafer 1

Increase in the number of dies per wafer = $0.1 * 84 = 8.4$

New number of dies per wafer = $84 + 8.4 = 92.4$

Increase in defects per unit area = $0.15 * 0.020 = 0.003$

New number of defects per unit area = $0.02 + 0.003 = 0.023$

New area per die = $176.71/92.4 = 1.912 \text{ cm}^2$

New yield = $1/(1+(0.023*1.912)/2)^2 = 0.957$

Wafer 2

Increase in the number of dies per wafer = $0.1 * 100 = 10$

New number of dies per wafer = $100 + 10 = 110$

Increase in defects per unit area = $0.15 * 0.031 = 0.005$

New number of defects per unit area = $0.031 + 0.005 = 0.036$

New area per die = $314.159/110 = 2.85 \text{ cm}^2$

New yield = $1/(1+(0.036*2.85)/2)^2 = 0.905$

1.10.4

Part 1 (for 0.92 technology)

yield = $1/(1+(\text{defects per unit area} * \text{dies per unit area}/2))^2$

$(1+(\text{defects per unit area} * \text{dies per unit area}/2))^2 = 1/\text{yield}$

$1+(\text{defects per unit area} * \text{dies per unit area}/2) = 1/\sqrt{\text{yield}}$

$\text{defects per unit area} * \text{dies per unit area}/2 = (1/\sqrt{\text{yield}}) - 1$

$\text{defects per unit area} = (2(1/\sqrt{\text{yield}}) - 1)/\text{dies per unit area}$

$200 \text{ mm}^2 = 2 \text{ cm}^2$

$\text{defects per unit area} = (2(1/\sqrt{0.92}) - 1)/2 = 0.042 \text{ cm}^2$

Part 2 (for 0.95 technology)

yield = $1/(1+(\text{defects per unit area} * \text{dies per unit area}/2))^2$

$(1+(\text{defects per unit area} * \text{dies per unit area}/2))^2 = 1/\text{yield}$

$1+(\text{defects per unit area} * \text{dies per unit area}/2) = 1/\sqrt{\text{yield}}$

$\text{defects per unit area} * \text{dies per unit area}/2 = (1/\sqrt{\text{yield}}) - 1$

$\text{defects per unit area} = (2(1/\sqrt{\text{yield}}) - 1)/\text{dies per unit area}$

$200 \text{ mm}^2 = 2 \text{ cm}^2$

$\text{defects per unit area} = (2(1/\sqrt{0.95}) - 1)/2 = 0.026 \text{ cm}^2$

1.14

$$\text{FP instructions} = 50 * 10^6$$

$$\text{INT instructions} = 110 * 10^6$$

$$\text{L/S instructions} = 80 * 10^6$$

$$\text{Branch instructions} = 16 * 10^6$$

Execution time = (Sum of the product of the instructions and their respective CPI number)/Clock rate

$$= (50 * 10^6 * 1 + 110 * 10^6 + (80 * 10^6 * 4) + (16 * 10^6 * 2)) / 2 * 10^9$$

$$= (50 * 10^6 + 110 * 10^6 + 320 * 10^6 + 32 * 10^6) / 2 * 10^9$$

$$= (512 * 10^6) / 2 * 10^9$$

$$= 256 * 10^{-3} = 0.256$$

1.14.1

If we want the program to run 2 times faster then we may have to reduce the execution time by half.

$$0.256/2 = 0.128$$

$$0.128 = (50x + 110 + 320 + 32) * 10^6 / 2 * 10^9$$

$$0.128 = (50x + 462) / 2 * 10^6$$

$$0.256 * 10^6 = 50x + 462$$

$$256 = 50x + 462$$

$$256 - 462 = 50x$$

$$-206 = 50x$$

$$-206/50 = x$$

$$-4.12 = x$$

Since CPI cannot be negative, we can say that it is not possible for this function to run 2 times faster.

1.14.2

If we want the program to run 2 times faster then we may have to reduce the execution time by half.

$$0.256/2 = 0.128$$

$$0.128 = (50 + 110 + 80x + 32) * 10^6 / 2 * 10^9$$

$$0.128 = (192 + 80x) * 10^6 / 2 * 10^9$$

$$0.128 = (192 + 80x) / 2 * 10^6$$

$$0.256 * 10^6 = 192 + 80x$$

$$256 = 192 + 80x$$

$$64 = 80x$$

$$64/80 = x$$

$$0.8 = x$$

Therefore, the new L/S instructions CPI must be 0.8 if we want the program to run 2 times faster.

1.14.3

New FP and INT CPI

$$1 * 0.4 = 0.4$$

$$1 - 0.4 = 0.6$$

Hence, 0.6 is the new FP and INT CPI

New Branch CPI

$$4 * 0.3 = 1.2$$

$$4 - 1.2 = 2.8$$

Hence, 2.8 is the new Branch CPI

New L/S CPI

$$2 * 0.3 = 0.6$$

$$2 - 0.6 = 1.4$$

Hence, 1.4 is the new L/S CPI.

$$\begin{aligned}\text{New execution time} &= ((50 * 0.6) + (110 * 0.6) + (80 * 2.8) + (16 * 1.4)) * 10^6 / 2 * 10^9 \\ &= (30 + 66 + 224 + 22.4) / 2 * 10^3 \\ &= 0.17\end{aligned}$$

In order to determine by how the execution time speeded we have to take the ratio of the previous execution time and the new execution time.

$$0.256 / 0.17 = 1.5$$

Hence, we can say that the execution time sped up by 1.5x.

$$\text{Percentage of time reduction} = ((1 - 0.17) / 0.256) * 100 = 33\%$$

Therefore, we can say that time reduced by 33%.

1.15

I have made the following table to answer this question:

p	$T_p(s)$	Actual Speed up	Actual / Ideal
2	54	1.9	0.95
4	29	3.4	0.85
8	16.5	6.1	0.763
16	10.3	9.8	0.613
32	7.1	14.00	0.439
64	5.6	18.00	0.28
128	4.9	21.00	0.16

Formulas Used

1) Time taken to run 1 processor = 100s

2) We know that the time taken to compute P processors is:

$$T_p = (100/p) + 4 \text{ (4 additional seconds of overhead per processor)}$$

3) Actual Speed up = $T_1/T_p = 100/(100/p + 4)$

4) Ideal speed up = p (since there is no overhead)